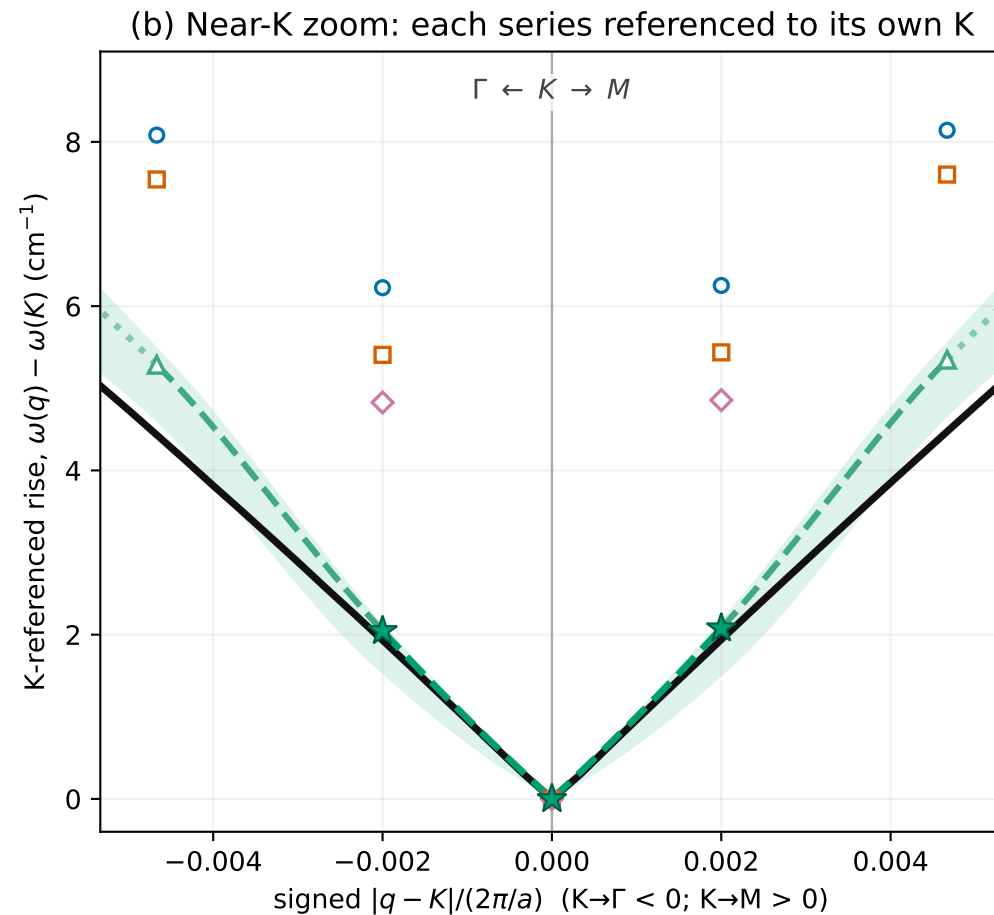
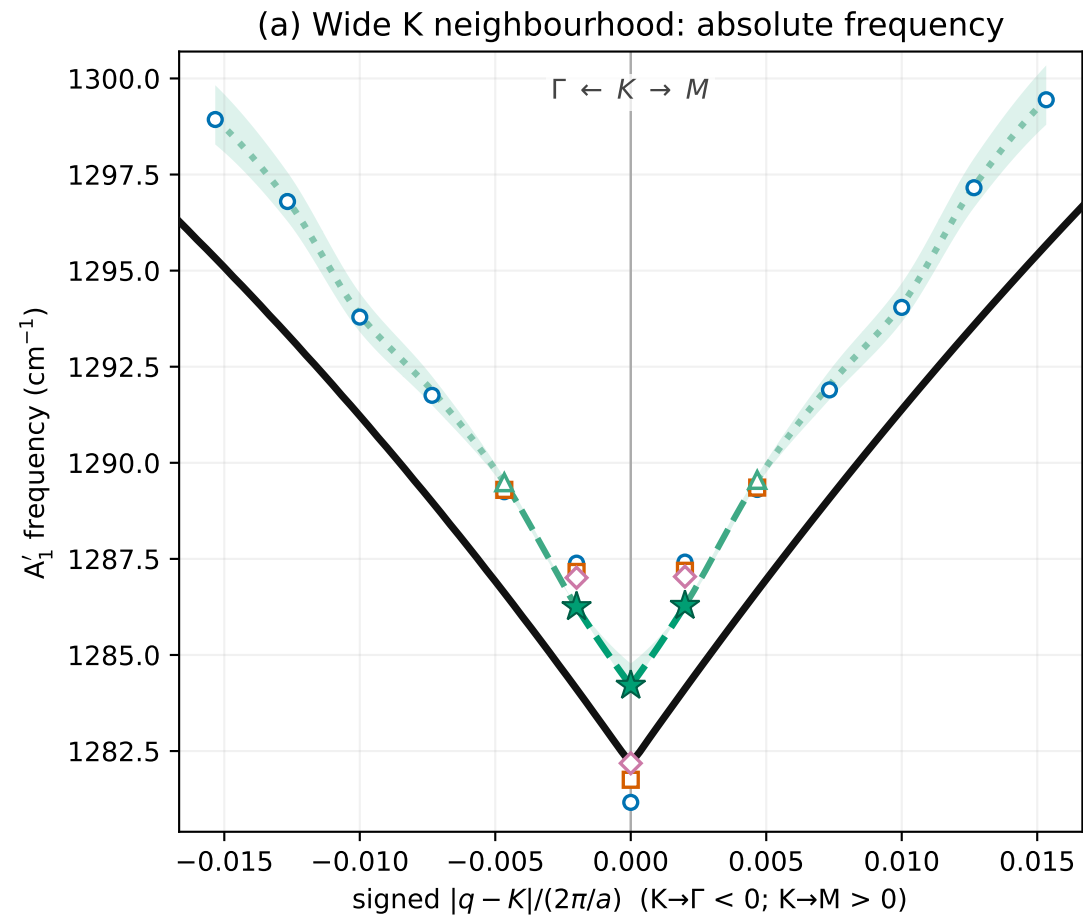


Graphene K-point A'_1 branch: wide zero-smearing extrapolation sketch



smearing/degauss = 0 Ry

- MLIP + full EPC LR
- - central DFPT $N \rightarrow \infty$ guide
- fit/support envelope (not a confidence interval)
- ★ three-grid fixed-q intercept
- △ two-grid $d=0.007$ intercept
- raw DFPT k192
- raw DFPT k240
- ◇ raw DFPT k288

Extrapolation support:
 $|d| \leq 0.003$: three electronic k grids
 $0.003 < |d| \leq 0.007$: two grids
 $0.007 < |d| \leq 0.023$: k192 plus a
decaying finite-grid correction (dotted).

The outer green curve is a provisional guide,
not a converged full DFPT dispersion.

Layout reference: Piscanec et al.,
Phys. Rev. Lett. 93, 185503 (2004), Fig. 2.